



DP-600 : Implement analytics solutions using Microsoft Fabric

Exercise 5 : Design and implement a dimensional model

- **Create a workspace**
- **Create a data warehouse**
- **Create the fact table**
- **Create the dimension tables**
- **Add table constraints**
- **Load sample data**
- **Query the star schema**
- **Implement SCD patterns**
- **Verify the design**



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Create a workspace

Create a workspace

Name *
dimensional model_5
✓ This name is available

Description
Describe this workspace

Domain ⓘ
Assign to a domain (optional)

[Learn more about workspace settings](#)

Workspace image
Upload
Reset

Advanced ^

Contact list * ⓘ
rsidvika (Owner)

Workspace type ⓘ
What a user can do in a workspace depends on the individual user license(s) assigned to that user and the workspace type. [Learn more](#)

Power BI only workspace types
Power BI workspaces only support the use of Power BI capabilities.

Apply **Cancel**

WS name: dimensional model_5

WS Type: Trial



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➤ Create a data warehouse

New Warehouse

Name *

ContosoDW

Location *



dimensional model_5



Advanced

Create

Cancel



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create the fact table

The screenshot displays the Microsoft Fabric SQL query editor interface. On the left, the 'Explorer' pane shows the 'ContosoDW' warehouse structure, including schemas, tables, and views. The 'f_Sales' table is highlighted. The main editor area shows a SQL query to create the 'f_Sales' table with the following columns and data types:

```
13 Create Table f_Sales
14 (
15     DateKey INT NOT NULL,
16     StoreKey INT NOT NULL,
17     ProductKey INT NOT NULL,
18     CustomerKey INT NOT NULL,
19     Quantity INT NOT NULL,
20     UnitPrice DECIMAL(10,2) NOT NULL,
21     SalesAmount DECIMAL(10,2) NOT NULL,
22     DiscountAmount DECIMAL(10,2) NOT NULL
23 );
24
25
26
```

The 'Messages' pane at the bottom shows the execution results:

```
2:16:17 PM Started executing on line 1
Statement ID: {125DDC4D-EBFB-4BC8-8F50-D63319252875}
2:16:19 PM Query execution time: 00:00:00.255 | Time to process and render results: 00:00:01.521 | Total duration: 00:00:01.776
```

The status bar at the bottom indicates 'AutoSave: On (Saved)', 'Succeeded (1 sec 776 ms)', and 'Copilot completions: Off'.



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➤ Create the dimension tables

The screenshot displays the Microsoft Fabric web interface. On the left, the 'Explorer' pane shows a tree view of the 'ContosoDW' workspace. Under 'Tables', the 'd_Date' table is expanded, listing columns: DateKey, FullDate, Year, Quarter, Month, MonthName, Day, DayofWeek, FiscalYear, FiscalQuarter, IsHoliday, and IsWeekday. The main pane shows 'SQL query 2' with a 'Run' button and a 'Preview' section. The SQL query is as follows:

```
1 create TABLE d_Date
2 (
3   DateKey INT NOT NULL,
4   FullDate DATE NOT NULL,
5   [Year] INT NOT NULL,
6   [Quarter] INT NOT NULL,
7   [Month] INT NOT NULL,
8   MonthName VARCHAR(10) NOT NULL ,
9   [Day] INT NOT NULL,
10  [DayofWeek] VARCHAR(10) NOT NULL,
11  FiscalYear INT NOT NULL,
12  FiscalQuarter INT NOT NULL,
13  IsHoliday BIT NOT NULL,
14  IsWeekday BIT NOT NULL
15 );
16
```

Below the query, the 'Messages' section shows the execution status:

- 2:26:43 PM: Started executing on line 1
- 2:26:45 PM: Statement ID: {8CC16CE6-30E4-4410-A32F-88CDF90CCEE}
- 2:26:45 PM: Query execution time: 00:00:00.249 | Time to process and render results: 00:00:01.476 | Total duration: 00:00:01.725

The bottom status bar indicates 'AutoSave: On (Saved)', 'Succeeded (1 sec 725 ms)', and 'Copilot completions: Off'.

TYPE-1 SCD : d_Date



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Create the dimension tables

The screenshot displays the Microsoft Fabric SQL query editor interface. On the left, the 'Explorer' pane shows a tree view of the database structure under 'ContosoDW'. The 'Tables' folder is expanded, showing a list of tables including 'd_Customer', 'd_Date', 'f_Sales', 'Views', 'Functions', 'Stored Procedures', and 'INFORMATION_SCHEMA'. The 'd_Customer' table is selected, and its columns are listed: 'CustomerKey', 'CustomerName', 'Segment', 'City', '[State]', 'Country', 'LoyaltyTier', and 'JoinDate'. The main editor area shows a SQL query for creating the 'd_Customer' table. The query is as follows:

```
1  
2  
3  
4 CREATE TABLE d_Customer  
5 (  
6     CustomerKey INT NOT NULL,  
7     CustomerName VARCHAR(50) NOT NULL,  
8     Segment VARCHAR(20) NOT NULL,  
9     City VARCHAR(50) NOT NULL,  
10    [State] VARCHAR(50) NOT NULL,  
11    Country VARCHAR(20) NOT NULL,  
12    LoyaltyTier VARCHAR(20) NOT NULL,  
13    JoinDate DATE NOT NULL  
14 );
```

Below the query editor, the 'Messages' pane shows the execution results. The first message indicates that the query started executing on line 1 at 2:32:54 PM. The second message shows the query execution time as 00:00:00.155, the time to process and render results as 00:00:00.914, and the total duration as 00:00:01.069.

TYPE-1 SCD : d_Customer



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Create the dimension tables

The screenshot displays the Microsoft Fabric web interface. On the left, the 'Explorer' pane shows a hierarchy of data assets under 'Warehouses'. A dimension table 'd_Store' is selected, showing its columns: StoreKey, StoreNaturalKey, StoreName, StoreType, City, [State], Country, Region, OpenDate, ValidFrom, ValidTo, and IsCurrent. The main pane shows 'SQL query 2' with a 'Run' button and a 'Preview' tab. The SQL query is a CREATE TABLE statement for 'd_Store' with the following schema:

```
CREATE TABLE d_Store
(
    StoreKey INT NOT NULL,
    StoreNaturalKey VARCHAR(10) NOT NULL,
    StoreName VARCHAR(50) NOT NULL,
    StoreType VARCHAR(20) NOT NULL,
    City VARCHAR(50) NOT NULL,
    [State] VARCHAR(50) NOT NULL,
    Country VARCHAR(50) NOT NULL,
    Region VARCHAR(50) NOT NULL,
    OpenDate DATE NOT NULL,
    ValidFrom DATE NOT NULL,
    ValidTo DATE NOT NULL,
    IsCurrent BIT NOT NULL
);
```

Below the query editor, the 'Messages' pane shows the execution results:

Time	Message
2:35:50 PM	Started executing on line 1 Statement ID: {BEFCB824-5DEB-4FD3-9A3A-46BAE0B76B55}
2:35:51 PM	Query execution time: 00:00:00.270 Time to process and render results: 00:00:00.990 Total duration: 00:00:01.260

The bottom status bar indicates 'AutoSave: On (Saved)', 'Succeeded (1 sec 260 ms)', and 'Copilot completions: Off'.

TYPE-2 SCD : d_Store



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Create the dimension tables

The screenshot displays the Microsoft Fabric SQL query editor interface. On the left, the 'Explorer' pane shows a tree view of the 'ContosoDW' database, with the 'd_Product' table selected. The main editor area shows the SQL query for creating the 'd_Product' table. The query is as follows:

```
1 CREATE TABLE d_Product
2 (
3     ProductKey INT NOT NULL,
4     ProductNaturalKey VARCHAR(10) NOT NULL,
5     ProductName VARCHAR(50) NOT NULL,
6     Brand VARCHAR(50) NOT NULL,
7     Subcategory VARCHAR(50) NOT NULL,
8     Category VARCHAR(50) NOT NULL,
9     UnitCost DECIMAL(10,2) NOT NULL,
10    ValidFrom DATE NOT NULL,
11    ValidTo DATE NOT NULL,
12    IsCurrent BIT NOT NULL
13 );
```

Below the query editor, the 'Messages' pane shows the execution results:

```
2:37:40 PM Started executing on line 1
Statement ID: {B6408BD2-3D8E-4092-AE9D-01507200FB17}
2:37:40 PM Query execution time: 00:00:00.170 | Time to process and render results: 00:00:00.670 | Total duration: 00:00:00.840
```

The bottom status bar indicates that the query was successful, with an execution time of 0 seconds and 840 milliseconds.

TYPE-2 SCD : d_Product



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Add table constraints

The screenshot shows the Microsoft Fabric SQL query editor interface. On the left, the Explorer pane displays the database structure: Schemas (dbo) > Tables (d_Customer, d_Date, d_Product, d_Store, f_Sales) > Views. The main editor area shows a SQL query with four ALTER TABLE statements, each adding a PRIMARY KEY NONCLUSTERED constraint to a specific column. The Messages pane at the bottom shows the execution results, indicating that the query executed successfully.

```
1 ALTER TABLE d_Date
2   ADD CONSTRAINT PK_d_Date PRIMARY KEY NONCLUSTERED (DateKey) NOT ENFORCED;
3
4 ALTER TABLE d_Store
5   ADD CONSTRAINT PK_d_Store PRIMARY KEY NONCLUSTERED (StoreKey) NOT ENFORCED;
6
7 ALTER TABLE d_Product
8   ADD CONSTRAINT PK_d_Product PRIMARY KEY NONCLUSTERED (ProductKey) NOT ENFORCED;
9
10 ALTER TABLE d_Customer
11   ADD CONSTRAINT PK_d_Customer PRIMARY KEY NONCLUSTERED (CustomerKey) NOT ENFORCED;
12
13
14
```

Messages Copy messages

Time	Message
2:44:03 PM	Started executing on line 1 Statement ID: {16C54709-B57A-41C7-865C-A93140C4A171} Statement ID: {8468D421-3F72-420B-A2C4-EF19C6F3A6B0} Statement ID: {A5C2FA37-FC62-45D3-B4E8-19BA2AC1FB67} Statement ID: {AE4A40B0-199D-4032-9398-C0ABB64D6955}
2:44:04 PM	Query execution time: 00:00:00.055 Time to process and render results: 00:00:00.732 Total duration: 00:00:00.787

AutoSave: On (Saved) Succeeded (0 sec 787 ms) Copilot completions: Off

primary keys :

DateKey,StoreKey, ProductKey, CustomerKey.



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Add table constraints

The screenshot shows the Microsoft Fabric SQL query editor interface. The top bar includes the 'Fabric' logo, the workspace name 'ContosoDW', and a search bar. The left sidebar contains navigation icons for Home, Workspaces, Copilot, OneLake catalog, Monitor, Real-Time, Workloads, and My workspace. The main editor area displays three tabs: 'SQL query 1', 'SQL query 2', and 'SQL query 3'. The active tab, 'SQL query 3', contains the following SQL code:

```
1  -- Add foreign keys to the fact table
2  ALTER TABLE f_Sales
3      ADD CONSTRAINT FK_Sales_Date FOREIGN KEY (DateKey)
4      REFERENCES d_Date(DateKey) NOT ENFORCED;
5
6  ALTER TABLE f_Sales
7      ADD CONSTRAINT FK_Sales_Store FOREIGN KEY (StoreKey)
8      REFERENCES d_Store(StoreKey) NOT ENFORCED;
9
10 ALTER TABLE f_Sales
11     ADD CONSTRAINT FK_Sales_Product FOREIGN KEY (ProductKey)
12     REFERENCES d_Product(ProductKey) NOT ENFORCED;
13
14 ALTER TABLE f_Sales
15     ADD CONSTRAINT FK_Sales_Customer FOREIGN KEY (CustomerKey)
16     REFERENCES d_Customer(CustomerKey) NOT ENFORCED;
```

Below the code editor is a 'Messages' section with a 'Copy messages' button. It shows a timestamp '2:46:09 PM' and a message: 'Started executing on line 1'. Below this message are five statement IDs: {B134937D-0A22-4618-B4C0-C7F1DC886780}, {04D9B807-45AD-4D7C-87BF-EDD96D6FF2B5}, {F57EA074-9F2F-4747-AD64-7A9CA3F41EBE}, {E5047E1D-8F77-48E0-8E66-EB8E4B9DEA15}, and {4408AE04-59FB-469B-86C4-3E842AFD10EE}. The bottom status bar indicates 'AutoSave: On (Last saved: 5m ago)', 'Succeeded (0 sec 997 ms)', and 'Copilot completions: Off'.



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➤ Load sample data

Home

Management

Help

Search

Power BI trial: 54 days left

🔔

⚙️

⬇️

?

👤

Analyze data with

Share

Warehouses

ContosoDW

Schemas

dbo

Tables

d_Custor

d_Date

d_Produc

d_Store

f_Sales

Views

Functions

Stored Pro...

INFORMATIO...

queryinsights

sys

Security

d_Customer

d_Date

d_Product

f_Sales

d_Store

Data preview - f_Sales

Showing 1000 rows

Search

	123	DateKey	123	StoreKey	123	ProductKey	123	CustomerKey	123	Quantity	e*	UnitPrice	e*	SalesAmount	e*	DiscountAmount
1	20260105	1	1	1	1	1	1500.00	1500.00	0.00							
2	20260105	1	3	1	2	35.00	70.00	5.00								
3	20260112	2	2	2	1	1100.00	1100.00	100.00								
4	20260112	2	4	2	3	8.00	24.00	0.00								
5	20260209	3	1	3	2	1500.00	3000.00	150.00								
6	20260209	3	5	3	1	22.00	22.00	0.00								
7	20260302	1	2	5	1	1100.00	1100.00	0.00								
8	20260302	4	3	4	4	35.00	140.00	10.00								
9	20260406	2	1	2	1	1500.00	1500.00	75.00								
10	20260504	3	4	3	5	8.00	40.00	0.00								

Copy SQL connection string

Succeeded (0 sec 928 ms)

Columns: 8 Rows: 10



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► Query the star schema

The screenshot displays the Microsoft Fabric SQL query editor interface. The left sidebar shows the 'Explorer' pane with the 'ContosoDW' database structure, including schemas, tables, and views. The main editor area shows a SQL query for 'SQL query 4' that joins the 'f_Sales' fact table with 'd_Store' and 'd_Customer' dimension tables. The query calculates total sales and transaction counts, grouped by region and customer segment. The 'Results' tab at the bottom shows the output of the query, which is a table with 4 columns and 4 rows of data.

```
1  
2  
3  
4 SELECT  
5     s.Region,  
6     c.Segment,  
7     SUM(f.SalesAmount) AS TotalSales,  
8     Count(*) AS TransactionCount  
9 FROM f_Sales f  
10 JOIN d_Store s ON f.StoreKey = s.StoreKey  
11 JOIN d_Customer c ON f.CustomerKey=c.CustomerKey  
12 Group By s.Region, c.Segment  
13 ORDER BY s.Region, c.Segment;
```

	ABC Region	ABC Segment	e* TotalSales	123 TransactionCount
1	Central	Premium	3062.00	3
2	East	Budget	140.00	1
3	West	Premium	1570.00	2
4	West	Standard	3724.00	4

analyzing sales by store region and customer segment



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➤ Implement SCD patterns

Simulate an SCD Type 2 change

Suppose the cost of the Mountain Bike Pro increases from \$1,200 to \$1,350 effective March 1, 2026. An SCD Type 2 change expires the current row and inserts a new version.

The screenshot shows the Microsoft Fabric SQL query editor interface. The query is designed to simulate an SCD Type 2 change for the 'Mountain Bike Pro' product. It selects data from the 'f_Sales' fact table, joining it with 'd_Date' and 'd_Product' dimension tables. The query filters for the product 'Mountain Bike Pro' and orders the results by the full date. The results pane shows three rows of data, where the third row (dated 2026-04-06) represents a new version of the product with a higher unit cost (\$1,350) and a later valid from date (2026-01-01), effectively replacing the previous version.

```
1 SELECT
2     d.FullDate,
3     p.ProductName,
4     p.UnitCost AS ProductCostVersion,
5     p.ValidFrom AS CostEffectiveDate,
6     f.Quantity,
7     f.SalesAmount
8 FROM f_Sales f
9 JOIN d_Date d ON f.DateKey = d.DateKey
10 JOIN d_Product p ON f.ProductKey = p.ProductKey
11 WHERE p.ProductNaturalKey = 'MB-PRO'
12 ORDER BY d.FullDate;
```

	FullDate	ABC ProductName	e* ProductCostVersion	CostEffectiveDate	123 Quantity	e* SalesAmount
1	2026-01-05	Mountain Bike Pro	1200.00	2026-01-01	1	1500.00
2	2026-02-09	Mountain Bike Pro	1200.00	2026-01-01	2	3000.00
3	2026-04-06	Mountain Bike Pro	1200.00	2026-01-01	1	1500.00



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► Implement SCD patterns

Simulate an SCD Type 1 change

The product name “Water Bottle” needs to be corrected to “Insulated Water Bottle”

The screenshot displays the Microsoft Fabric SQL query editor interface. The top bar shows the Explorer pane on the left with a tree view containing 'Workspaces', 'Copilot', 'OneLake catalog', 'Monitor', 'Real-Time', 'Workloads', 'dimensional model_5', and 'My workspace'. The main editor area shows a SQL query for 'SQL query 5' with the following code:

```
1
2
3 UPDATE d_Product
4 SET ProductName='Insulated Water Bottle'
5 WHERE ProductNaturalKey='WB-STD';
6
7 SELECT ProductKey, ProductNaturalKey, ProductName, UnitCost, ValidFrom, ValidTo, IsCurrent
8 FROM d_Product
9 ORDER BY ProductNaturalKey, ValidFrom;
```

Below the query editor, the 'Results' tab is active, displaying a table with 6 rows and 9 columns. The table data is as follows:

	ProductKey	ProductNaturalKey	ProductName	UnitCost	ValidFrom	ValidTo	IsCurrent
1	3	HL-STD	Cycling Helmet	25.00	2026-01-01	9999-12-31	1
2	5	LK-STD	Bike Lock	15.00	2026-01-01	9999-12-31	1
3	1	MB-PRO	Mountain Bike Pro	1200.00	2026-01-01	2026-03-01	0
4	2	RB-ELT	Road Bike Elite	900.00	2026-01-01	9999-12-31	1
5	4	WB-STD	Insulated Water Bottle	5.00	2026-01-01	9999-12-31	1
6	6	mb-pro	Mountain Bike Pro	1350.00	2026-03-01	9999-12-31	1

The status bar at the bottom indicates 'AutoSave: On (Saved)', 'Succeeded (1 sec 513 ms)', and 'Copilot completions: Off'. The bottom right corner shows 'Columns: 7 Rows: 6'.



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Verify the design

Home Management Help

Analyze data with Share

Explorer

d_Customer d_Date d_Product f_Sales d_Store SQL query 4 SQL query 5

Run Save as Save as view Create rule Explain query Fix query errors

```
1 SELECT
2     d.FullDate,
3     d.[Year],
4     d.MonthName,
5     s.StoreName,
6     s.Region,
7     p.ProductName,
8     p.Category,
9     c.CustomerName,
10    c.Segment,
11    f.Quantity,
12    f.UnitPrice,
13    f.SalesAmount,
14    f.DiscountAmount
15 FROM f_Sales f
16 JOIN d_Date d ON f.DateKey = d.DateKey
17 JOIN d_Store s ON f.StoreKey = s.StoreKey
18 JOIN d_Product p ON f.ProductKey = p.ProductKey
19 JOIN d_Customer c ON f.CustomerKey = c.CustomerKey
20 ORDER BY d.FullDate, s.StoreName;
```

Home Management Help

Analyze data with Share

Explorer

d_Customer d_Date d_Product f_Sales d_Store SQL query 4 SQL query 5

Run Save as Save as view Create rule Explain query Fix query errors

Messages Results

Search

	FullDate	123 Year	ABC MonthNa...	ABC StoreName	ABC Region	ABC ProductN...	ABC Category	ABC Customer...	ABC Segment	123 Quantity	e ^x UnitPrice	e ^x SalesAmo...	e ^x Discount...
1	2026-01-05	2026	January	Contoso Dow...	West	Mountain Bike...	Bikes	Jordan Rivera	Premium	1	1500.00	1500.00	0.00
2	2026-01-05	2026	January	Contoso Dow...	West	Cycling Helmet	Accessories	Jordan Rivera	Premium	2	35.00	70.00	5.00
3	2026-01-12	2026	January	Contoso Mall	West	Road Bike Elite	Bikes	Alex Chen	Standard	1	1100.00	1100.00	100.00
4	2026-01-12	2026	January	Contoso Mall	West	Insulated Wat...	Accessories	Alex Chen	Standard	3	8.00	24.00	0.00
5	2026-02-09	2026	February	Contoso Central	Central	Mountain Bike...	Bikes	Sam Patel	Premium	2	1500.00	3000.00	150.00
6	2026-02-09	2026	February	Contoso Central	Central	Bike Lock	Accessories	Sam Patel	Premium	1	22.00	22.00	0.00
7	2026-03-02	2026	March	Contoso Dow...	West	Road Bike Elite	Bikes	Morgan Lee	Standard	1	1100.00	1100.00	0.00
8	2026-03-02	2026	March	Contoso Plaza	East	Cycling Helmet	Accessories	Taylor Kim	Budget	4	35.00	140.00	10.00
9	2026-04-06	2026	April	Contoso Mall	West	Mountain Bike...	Bikes	Alex Chen	Standard	1	1500.00	1500.00	75.00
10	2026-05-04	2026	May	Contoso Central	Central	Insulated Wat...	Accessories	Sam Patel	Premium	5	8.00	40.00	0.00
11	2026-05-04	2026	May	Contoso Dow...	West	Mountain Bike...	Bikes	Morgan Lee	Standard	1	1500.00	1500.00	0.00



/Julurishashi



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Reference

